

Activity 3

Week 4

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Activity: Orchestrating Microservices Containers

Instructions

1. Work individually. This assignment is due Monday September 7th, 2026 no later than 11:00 pm (23:00) on mycourseville (pdf only)
2. *NOTE:* you must run grader as I will mainly use the results from grader.
3. You will create docker-compose file to orchestrate the following components:
 - a) `todo-webapp` (`src=release-3.1`) - map port: container 3000 to host 3000
 - b) `todo` (`src=release-3`) - set name to `todo-service`
 - c) `todo-notification` (`src=release-1.1`) - set name to `notification-service`
 - d) API Gateway (using NGINX) - map port: container 8000 to host 8000 - set name to `api-gateway`
 - e) Network name: `todo-net`
4. `todo`, `todo-notification`, API Gateway must be all connected to the same internal network.
5. You must modify nginx configuration file as followed:
 - a) Listen to port 8000.
 - b) Point proxy from `/todo/` to <http://todo-service:8000/>
 - c) Point proxy from `/notification/` to <http://notification-service:9000/>
6. Prepare a set of slides in PDF containing the following information.
 - a) Your `docker-compose.yml`.
 - b) Screenshot when you run `docker-compose up`.
 - c) Screenshot after all containers are running and you use command: `docker ps -a`.
 - d) Screenshot of browser connecting to `todo-webapp` with some todo items.

YOU MUST ALSO USING GRADER (URL is in MCV's assignment)

The answers and solutions are provided in the next page...

Problem 1. Your docker-compose.yml.

Solution

Here's the docker-compose.yml file:

```
1  name: activity-03
2
3  services:
4    todo-webapp:
5      build:
6        context: ./src/todo-webapp
7      container_name: webapp
8      ports:
9        - "3000:3000"
10     stdin_open: true
11     tty: true
12     networks:
13       - todo-net
14     depends_on:
15       - api-gateway
16
17     todo:
18       build:
19         context: ./src/todo
20       container_name: todo-service
21       environment:
22         REDIS_HOST: redis
23         REDIS_PORT: "6379"
24         NOTIFICATION_HOST: notification-service
25         NOTIFICATION_PORT: "9000"
26       networks:
27         - todo-net
28       depends_on:
29         - redis
30         - todo-notification
31
32     todo-notification:
33       build:
34         context: ./src/todo-notification
35       container_name: notification-service
36       environment:
37         REDIS_HOST: redis
38         REDIS_PORT: "6379"
39       networks:
40         - todo-net
41       depends_on:
42         - redis
43
44     api-gateway:
45       image: nginx:alpine
46       container_name: api-gateway
47       ports:
48         - "8000:8000"
49       volumes:
50         - ./nginx/default.conf:/etc/nginx/conf.d/default.conf:ro
51       networks:
52         - todo-net
53       depends_on:
54         - todo
55         - todo-notification
56
57     redis:
58       image: redis:7-alpine
59       container_name: redis
60       networks:
61         - todo-net
```

```

62
63 networks:
64   todo-net:
65     name: todo-net
66     driver: bridge

```

Problem 2. Screenshot when you run docker-compose up.

Solution

```

nodahtap@LAPTOP-UP5JACSH:~/study/cedt-3-1/2110506-SDS/activity-03/code$ docker compose up
[+] up 6/6
✓ Network todo-net          Created
✓ Container redis           Created
✓ Container notification-service Created
✓ Container todo-service    Created
✓ Container api-gateway     Created
✓ Container webapp          Created
Attaching to api-gateway, notification-service, redis, todo-service, webapp

```

Figure 1: Output of docker compose up.

Problem 3. Screenshot after all containers are running and you use command: docker ps -a.

Solution

```

nodahtap@LAPTOP-UP5JACSH:~/study/cedt-3-1/2110506-SDS/activity-03/code$ docker ps -a
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                               NAMES
feec161539a4   redis:7-alpine "docker-entrypoint.s..." 5 minutes ago Up 5 minutes 6379/tcp                               redis
52208d634985   84e2664cbac2  "docker-entrypoint.s..." 9 minutes ago Up 9 minutes 0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp webapp
4e4398388867   nginx:alpine  "/docker-entrypoint..." 9 minutes ago Up 9 minutes 80/tcp, 0.0.0.0:8000->8000/tcp, [::]:8000->8000/tcp api-gateway
62d00eb66c84   0579ff462068  "/bin/sh -c 'python ..." 9 minutes ago Up 9 minutes 8000/tcp                             todo-service
09efa7f4fa82   cd4b30c3caa7  "/bin/sh -c 'python ..." 9 minutes ago Up 9 minutes 9000/tcp                             notification-service

```

Figure 2: Output of docker ps -a.

Problem 4. Screenshot of browser connecting to todo-webapp with some todo items.

Solution

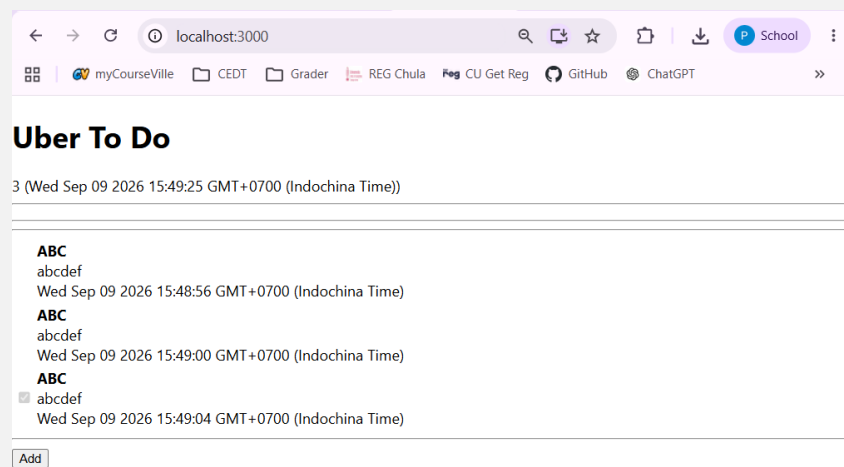


Figure 3: Screenshot of browser connecting to todo-webapp with some todo items.

Problem 5. Screenshot of the results after running the GRADER.

Solution

```
nodahttap@LAPTOP-UP5JACSH:~/study/cedt-3-1/2110506-SDS/activity-03/code$ ./activity3-linux-CEDT
2026/09/09 15:43:21 Current Timestamp: 2026-09-09T15:43:21+07:00
2026/09/09 15:43:21 ✓ / Network exists.
2026/09/09 15:43:22 ✓ / Container webapp exists.
2026/09/09 15:43:22 ✓ / Container todo-service exists.
2026/09/09 15:43:22 ✓ / Container notification-service exists.
2026/09/09 15:43:22 ✓ / Container redis exists.
2026/09/09 15:43:22 ✓ / Container api-gateway exists.
2026/09/09 15:43:22 ✓ / All specified containers are running.
2026/09/09 15:43:22 ✓ / Docker compose is running.
2026/09/09 15:43:22 ✓ / URL http://localhost:3000 returns 200.
2026/09/09 15:43:22 ✓ / Todo app is working.
2026/09/09 15:43:22 ✓ / URL http://localhost:8000 returns 404.
2026/09/09 15:43:22 ✓ / URL http://localhost:8000/todo returns 200.
2026/09/09 15:43:22 ✓ / URL http://localhost:8000/notification returns 200.
2026/09/09 15:43:22 ✓ / Notification-service found with api-gateway.
2026/09/09 15:43:22 Result: true
2026/09/09 15:43:22 🎉 Looks good! Please enter your StudentID and Full name below
👉 StudentID: 6733172621
👉 Full Name (TH): พัดชล เพ่งพินิจ
2026/09/09 15:43:33 🎉 Successfully submitted. Your lucky number is 21668149287704473
2026/09/09 15:43:33 🎉🎉🎉 Congratulations! You have completed the activity 🎉🎉🎉
2026/09/09 15:43:33 ⚠️ Don't forget! you still need to submit your assignment via MyCourseVille ⚠️
```

Figure 4: Screenshot of the results after running the GRADER.